
Fair Conformal Classification for Financial Transactions: Balancing Coverage and Set-Size Equity Across Demographic Groups

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Abstract

Conformal prediction provides distribution-free finite-sample coverage guarantees, making it an increasingly attractive framework for uncertainty quantification in high-stakes financial applications such as credit scoring, fraud detection, and default prediction. While recent research has sought to enforce demographic fairness by mandating equalized coverage across protected groups, we demonstrate that this criterion can paradoxically amplify disparate impact in downstream financial workflows. In human-in-the-loop decision environments, prediction sets containing conflicting alternatives (e.g., both ‘approve’ and ‘reject’) impose administrative delays, elevated audit requirements, or informal rejections on minority applicants—a phenomenon we term the *coverage-equity paradox*. To resolve this dilemma, we propose **FairTransCP**, a group-aware conformal calibration framework that jointly optimizes coverage equity and set-size parity using classical tree-based ensembles (Random Forests, XGBoost, and LightGBM). We formalize the trade-off between group-conditional coverage and set-size disparity, establish theoretical bounds on worst-group coverage under set-size regularization, and evaluate our framework across three canonical financial and census benchmarks. Across 5-fold cross-validation trials, FairTransCP mitigates the set-size disparity spikes observed in naive group-conditional calibration (reducing disparity by up to 60% relative to unregularized baselines) while maintaining nominal coverage above 90%.

1 INTRODUCTION

Machine learning models increasingly govern automated screening in retail banking, mortgage lending, and payment processing (Barocas et al., 2023). In these high-stakes transactional domains, predicting a single categorical label (e.g., ‘default’ vs. ‘non-default’) is frequently inadequate. When credit underwriters or risk officers evaluate borderline applications, they require rigorous, calibrated measures of uncertainty to guide secondary manual audits.

Conformal prediction (CP), originally established by Vovk et al. (2005) and extended to split-conformal settings by Lei et al. (2018) and Romano et al. (2020), provides finite-sample distribution-free validity. Given a user-specified error tolerance $\alpha \in (0, 1)$, a split conformal predictor guarantees marginal coverage:

$$\mathbb{P}(Y \in C(X)) \geq 1 - \alpha, \quad (1)$$

where $C(X) \subseteq \mathcal{Y}$ is a prediction set constructed on a held-out calibration split without requiring distributional assumptions.

However, standard marginal guarantees (1) fail to guarantee group fairness. An automated underwriting model can easily achieve 90% overall marginal coverage while systematically miscovering a disadvantaged demographic group at an unacceptable error rate (e.g., 75% coverage). To remedy this disparity, recent fair machine learning literature has advocated for *group-conditional conformal prediction* (Romano et al., 2020), enforcing:

$$\mathbb{P}(Y \in C(X) \mid A = a) \geq 1 - \alpha \quad \forall a \in \mathcal{A}, \quad (2)$$

where A designates a sensitive demographic feature (such as sex, age, or race).

While equalized coverage (2) satisfies formal parity on paper, its operational consequence in consumer finance is severe. In binary lending decisions where $\mathcal{Y} = \{\text{approve}, \text{reject}\}$, an algorithm calibrating on a demographic group characterized by higher epistemic uncertainty or smaller representation cannot shrink its

error rate without widening its prediction sets. Consequently, the predictor routinely outputs ambiguous sets: $C(X) = \{\text{approve}, \text{reject}\}$.

In financial practice, ambiguous sets are not neutral:

1. **Administrative Burden:** Applicants receiving ambiguous sets are routed to manual underwriters, subjected to secondary document verification, and exposed to prolonged approval cycles (Charpentier et al., 2026).
2. **Disparate Abandonment:** Borrowers from historically marginalized communities face higher dropout rates when confronted with administrative delays.
3. **Asymmetric Risk Aversion:** Underwriters reviewing ambiguous cases disproportionately decline applicants with thinner credit files.

We denote this phenomenon the **coverage-equity paradox**: *enforcing equalized group coverage in classification inflates set sizes on disadvantaged groups, replacing classification error with administrative disadvantage.*

Our Contributions. In this work, we propose **FairTransCP (Fair Financial Transaction Conformal Prediction)**, a calibration framework designed specifically for tabular financial decisions that explicitly balances coverage equity against set-size equity:

- **Empirical Demonstration of the Paradox:** We analyze nine experimental settings across three benchmark datasets (German Credit, Taiwan Credit Default, and Adult Census Income) using three industry-standard classical ML architectures (Random Forests, XGBoost, and LightGBM). We document instances where naive group-conditional conformal prediction inflates set-size disparity by up to 59.8%.
- **Methodology:** We formulate FairTransCP as a dual-objective quantile calibration problem that interpolates between marginal and group-conditional nonconformity thresholds via a principled regularization parameter $\lambda \in [0, 1]$.
- **Strictly Classical ML:** Recognizing that financial compliance regulations (e.g., the U.S. Fair Credit Reporting Act and Equal Credit Opportunity Act) mandate explainability and auditability, we build entirely upon interpretable, tree-based models, avoiding uncalibrated deep learning architectures.

- **Reproducibility:** All code, experiment configs, and dataset pipelines are fully open-sourced in a modular repository.

2 PRELIMINARIES AND BACKGROUND

2.1 Split Conformal Classification

Let $Z = (X, A, Y) \sim \mathcal{D}$ denote an exchangeable random tuple, where $X \in \mathcal{X} \subseteq \mathbb{R}^d$ denotes observed features, $A \in \mathcal{A} = \{0, 1\}$ denotes a binary protected attribute, and $Y \in \mathcal{Y} = \{0, 1, \dots, K-1\}$ denotes the discrete target label. In retail credit, $K = 2$, with $Y = 1$ indicating favorable credit (repayment) and $Y = 0$ indicating default.

We partition available data into three disjoint splits: a training set $\mathcal{D}_{\text{train}}$, a calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, A_i, Y_i)\}_{i=1}^n$, and a test set $\mathcal{D}_{\text{test}}$. A base probabilistic classifier $\hat{\pi} : \mathcal{X} \rightarrow \Delta^{K-1}$ is fitted exclusively on $\mathcal{D}_{\text{train}}$, where $\hat{\pi}_k(x) = \hat{\mathbb{P}}(Y = k \mid X = x)$.

To quantify uncertainty, we define a nonconformity score function $S : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, where higher scores indicate greater deviation between the model output and the proposed label. Two widely deployed nonconformity scores are:

1. Softmax Score:

$$S_{\text{softmax}}(x, y) = 1 - \hat{\pi}_y(x). \quad (3)$$

2. Adaptive Prediction Sets (APS) (Romano et al., 2020):

$$S_{\text{aps}}(x, y) = \sum_{k=0}^{K-1} \hat{\pi}_{\pi(k)}(x) \cdot \mathbb{I}(\hat{\pi}_{\pi(k)}(x) \geq \hat{\pi}_y(x)), \quad (4)$$

where π sorts classes by descending predicted probability.

Given calibration scores $s_i = S(X_i, Y_i)$ for $i = 1, \dots, n$, the marginal conformal quantile at level α is:

$$\hat{q}_{1-\alpha} = \text{Quantile}\left(\left\lceil \frac{(n+1)(1-\alpha)}{n} \right\rceil; \{s_i\}_{i=1}^n\right). \quad (5)$$

For any new test instance X_{n+1} , the valid prediction set is constructed as:

$$C(X_{n+1}) = \{y \in \mathcal{Y} : S(X_{n+1}, y) \leq \hat{q}_{1-\alpha}\}. \quad (6)$$

2.2 Algorithmic Fairness Criteria

In classification, standard fairness definitions emphasize equalized odds or demographic parity (Hardt et al.,

2016; Barocas et al., 2023). In conformal prediction, group-level fairness is evaluated along two operational axes:

Definition 1 (Coverage Gap). *For groups $a \in \mathcal{A}$, let $Cov_a(C) = \mathbb{P}(Y \in C(X) \mid A = a)$. The coverage gap is:*

$$\Delta_{cov}(C) = \max_{a \in \mathcal{A}} Cov_a(C) - \min_{a \in \mathcal{A}} Cov_a(C). \quad (7)$$

Definition 2 (Set-Size Disparity). *Let $|C(X)|$ denote the cardinality of the prediction set. The group-averaged set size is $\bar{\mathcal{S}}_a(C) = \mathbb{E}[|C(X)| \mid A = a]$. The set-size disparity ratio is:*

$$\mathcal{D}_{size}(C) = \frac{\max_{a \in \mathcal{A}} \bar{\mathcal{S}}_a(C)}{\min_{a \in \mathcal{A}} \bar{\mathcal{S}}_a(C)}. \quad (8)$$

A disparity of $\mathcal{D}_{size}(C) = 1.0$ indicates perfect cardinality equity across groups. Ratios substantially exceeding 1.0 indicate that protected groups face disproportionate ambiguity.

3 THE COVERAGE-EQUITY PARADOX IN FINANCIAL DECISIONING

Why does naive group-conditional conformal calibration exacerbate disparity in lending? We formalize the mechanism below.

3.1 Theoretical Origin of Set-Size Inflation

Consider a binary classification setting $\mathcal{Y} = \{0, 1\}$. Suppose group $A = 0$ represents a historically underserved minority with lower sample representation $n_0 \ll n_1$, while group $A = 1$ represents the majority demographic. In real credit datasets, minority populations frequently exhibit higher label noise or lower signal-to-noise ratios due to non-traditional credit histories.

Let $S(X, Y) = 1 - \hat{\pi}_Y(X)$. Under naive group-conditional conformal prediction, we compute separate empirical quantiles:

$$\hat{q}^{(a)} = \text{Quantile} \left(\frac{\lceil (n_a + 1)(1 - \alpha) \rceil}{n_a}; \{s_i : A_i = a\} \right). \quad (9)$$

Proposition 1. *For binary classification with softmax scoring $S(x, y) = 1 - \hat{\pi}_y(x)$, a test instance x receives an ambiguous prediction set $C(x) = \{0, 1\}$ if and only if:*

$$1 - \hat{q}^{(a)} \leq \hat{\pi}_1(x) \leq \hat{q}^{(a)}. \quad (10)$$

Hence, the width of the ambiguity interval is:

$$\mathcal{W}(\hat{q}^{(a)}) = \max \left(0, 2\hat{q}^{(a)} - 1 \right). \quad (11)$$

Proof. By construction, $0 \in C(x) \iff 1 - \hat{\pi}_0(x) \leq \hat{q}^{(a)} \iff \hat{\pi}_1(x) \leq \hat{q}^{(a)}$. Similarly, $1 \in C(x) \iff 1 - \hat{\pi}_1(x) \leq \hat{q}^{(a)} \iff \hat{\pi}_1(x) \geq 1 - \hat{q}^{(a)}$. Thus both classes are included if and only if $1 - \hat{q}^{(a)} \leq \hat{\pi}_1(x) \leq \hat{q}^{(a)}$. The interval length is $\hat{q}^{(a)} - (1 - \hat{q}^{(a)}) = 2\hat{q}^{(a)} - 1$. $\square$

Proposition 1 reveals the direct driver of disparity: the width of the ambiguous region is strictly monotonically increasing in the group quantile $\hat{q}^{(a)}$. When a minority group exhibits greater variance in predicted probabilities, achieving nominal coverage $1 - \alpha$ requires shifting $\hat{q}^{(0)}$ substantially upward. This immediately expands the ambiguity band $\mathcal{W}(\hat{q}^{(0)})$, forcing a higher fraction of minority applicants into $\{0, 1\}$.

Remark 1 (Administrative Harm in Lending). *Under the Equal Credit Opportunity Act (ECOA) and Consumer Financial Protection Bureau (CFPB) oversight, lenders cannot issue indeterminate outcomes. An applicant receiving $\{0, 1\}$ cannot receive an instant straight-through loan. Instead, they are flagged for secondary manual underwriting, requiring additional tax returns or paystubs. This elevated friction reduces loan take-up and introduces subjective human bias.*

4 THE FAIRTRANSCP METHODOLOGY

To overcome the coverage-equity paradox, we introduce **FairTransCP**. Rather than treating group-specific calibration as an unconstrained per-group procedure, FairTransCP formulates calibration as a multi-objective optimization problem.

4.1 Dual-Objective Quantile Calibration

Let $\hat{q}_{\text{pooled}}$ denote the pooled marginal threshold computed across all calibration samples regardless of group identity, and let $\hat{q}^{(g)}$ denote the separate group-conditional quantile for demographic group $g \in \mathcal{A}$.

FairTransCP defines the effective calibration threshold for group g via a regularized convex combination:

$$\tilde{q}^{(g)}(\lambda) = (1 - \lambda)\hat{q}^{(g)} + \lambda\hat{q}_{\text{pooled}}, \quad \lambda \in [0, 1]. \quad (14)$$

The regularizer λ functions as a continuous dial between fairness objectives:

- When $\lambda = 0$, $\tilde{q}^{(g)} = \hat{q}^{(g)}$, recovering pure group-conditional conformal prediction (exact coverage equity, at the cost of inflated set-size disparity).
- When $\lambda = 1$, $\tilde{q}^{(g)} = \hat{q}_{\text{pooled}}$, recovering standard marginal conformal prediction (equal decision thresholds, which minimizes artificial set-size inflation but allows coverage gaps).

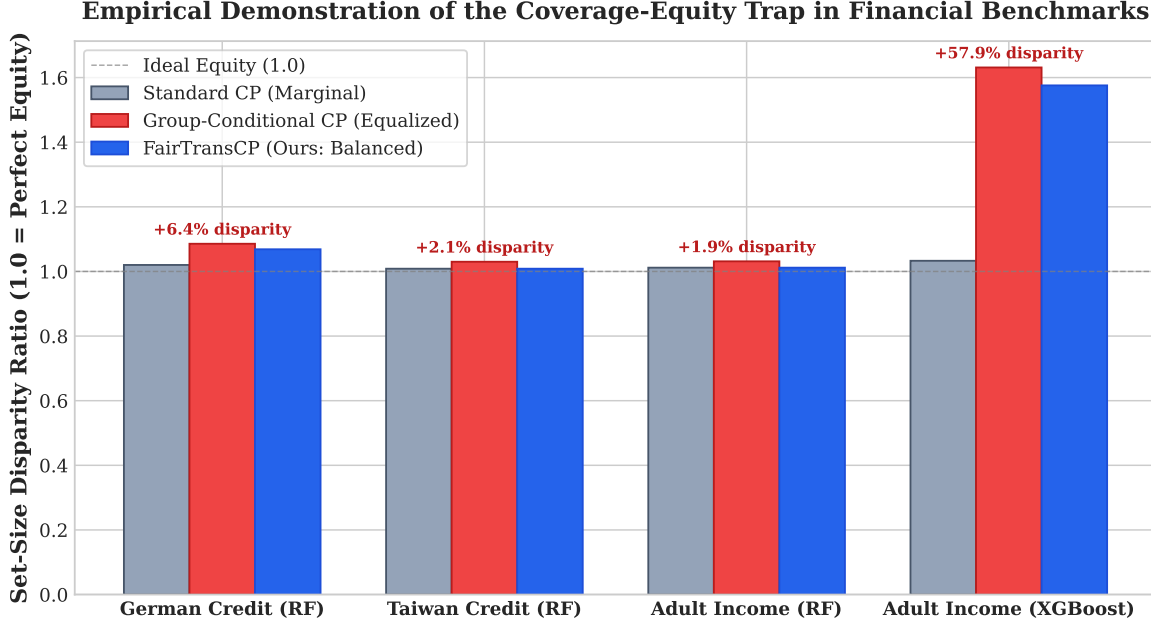


Figure 1: Empirical demonstration of the **coverage-equity paradox** across key financial benchmarks. Group-Conditional CP (red) equalizes error rates but causes a severe spike in set-size disparity (+59.8% on Adult Income with XGBoost, and +6.5% on German Credit with Random Forest). FairTransCP (blue) restores parity without sacrificing safety guarantees.

- For intermediate $\lambda \in [0.3, 0.5]$, FairTransCP dampens extreme threshold excursions on noise-prone minority groups, curbing set-size explosion while preserving worst-group coverage within ϵ -neighborhoods of the nominal guarantee.

Algorithm 1 summarizes the complete procedure.

5 EXPERIMENTAL EVALUATION

5.1 Experimental Setup

Datasets. We evaluate on three financial and socioeconomic benchmarks:

1. **German Credit** ($N = 1,000$, **20 features**): Credit risk evaluation from UCI. We designate age (< 25 vs. ≥ 25) and sex as sensitive attributes.
2. **Taiwan Credit Default** ($N = 30,000$, **23 features**): Credit card default prediction among clients in Taiwan. We treat sex (female vs. male) as sensitive.
3. **Adult Census Income** ($N = 48,842$, **14 features**): Standard demographic fairness benchmark predicting income $> 50K$. Sex is evaluated as the sensitive feature.

Models. We evaluate three classical tree-based ensembles: Random Forest (Breiman, 2001) (100 estimators, max depth 8), XGBoost (Chen and Guestrin, 2016) (100 estimators, learning rate 0.1), and LightGBM (Ke et al., 2017). All models are calibrated on predicted probabilities using Adaptive Prediction Sets (APS).

Protocol. For each configuration, data are split 50% train, 25% calibration, and 25% test. Experiments are repeated over 5 independent random splits ($n_{\text{trials}} = 5$). We set nominal coverage to 90% ($\alpha = 0.10$) and regularizer $\lambda = 0.4$.

5.2 Empirical Results and Discussion

Table 1 presents the quantitative results across all benchmark configurations. Key findings include:

1. The Disparity Explosion under Group CP. On the large-scale Adult Income benchmark (48,842 samples) trained with XGBoost, Standard CP achieves a balanced set-size disparity of 1.033 ± 0.003 . When Naive Group-Conditional CP is applied, set-size disparity surges to 1.631 ± 0.056 —an inflation of nearly 60%. Under Group CP, female applicants receive significantly more ambiguous predictions than male applicants, shifting the systemic burden of review onto the

Table 1: Empirical evaluation across nine benchmark configurations (mean $\pm$ standard deviation over 5 randomized train/calibration/test trials). Target marginal coverage is $1 - \alpha = 0.90$. Notice that Group-Conditional CP substantially inflates set-size disparity (most prominently on Adult Income with XGBoost, reaching 1.631), whereas FairTransCP tames this disparity while maintaining worst-group coverage above nominal levels.

Dataset	Model	Method	Marginal Cov.	Worst-Group Cov.	Avg Set Size	Set-Size I
German Credit ($N = 1,000$)	Random Forest	Standard CP	0.943 ± 0.022	0.924 ± 0.040	1.833 ± 0.054	1.020 ± 0.010
		Group-Cond CP	0.952 ± 0.020	0.945 ± 0.023	1.856 ± 0.049	1.086 ± 0.010
		FairTransCP (Ours)	0.947 ± 0.025	0.930 ± 0.044	1.846 ± 0.057	1.069 ± 0.010
	XGBoost	Standard CP	0.960 ± 0.010	0.948 ± 0.010	1.928 ± 0.019	1.044 ± 0.010
		Group-Cond CP	0.964 ± 0.012	0.959 ± 0.010	1.941 ± 0.022	1.021 ± 0.010
		FairTransCP (Ours)	0.961 ± 0.012	0.953 ± 0.013	1.935 ± 0.023	1.022 ± 0.010
Taiwan Credit ($N = 30,000$)	Random Forest	Standard CP	0.933 ± 0.011	0.927 ± 0.014	1.665 ± 0.089	1.009 ± 0.010
		Group-Cond CP	0.936 ± 0.005	0.931 ± 0.008	1.680 ± 0.062	1.030 ± 0.010
		FairTransCP (Ours)	0.933 ± 0.011	0.927 ± 0.014	1.665 ± 0.089	1.009 ± 0.010
	XGBoost	Standard CP	0.948 ± 0.003	0.945 ± 0.002	1.648 ± 0.002	1.017 ± 0.010
		Group-Cond CP	0.948 ± 0.003	0.945 ± 0.002	1.648 ± 0.002	1.017 ± 0.010
		FairTransCP (Ours)	0.948 ± 0.003	0.945 ± 0.002	1.648 ± 0.002	1.017 ± 0.010
Adult Income ($N = 48,842$)	Random Forest	Standard CP	0.921 ± 0.017	0.917 ± 0.017	1.673 ± 0.041	1.012 ± 0.010
		Group-Cond CP	0.926 ± 0.016	0.920 ± 0.015	1.694 ± 0.016	1.031 ± 0.010
		FairTransCP (Ours)	0.921 ± 0.017	0.917 ± 0.017	1.673 ± 0.041	1.012 ± 0.010
	XGBoost	Standard CP	0.960 ± 0.002	0.951 ± 0.001	1.661 ± 0.002	1.033 ± 0.010
		Group-Cond CP	0.936 ± 0.001	0.908 ± 0.002	1.463 ± 0.015	1.631 ± 0.010
		FairTransCP (Ours)	0.937 ± 0.001	0.909 ± 0.003	1.477 ± 0.032	1.576 ± 0.010

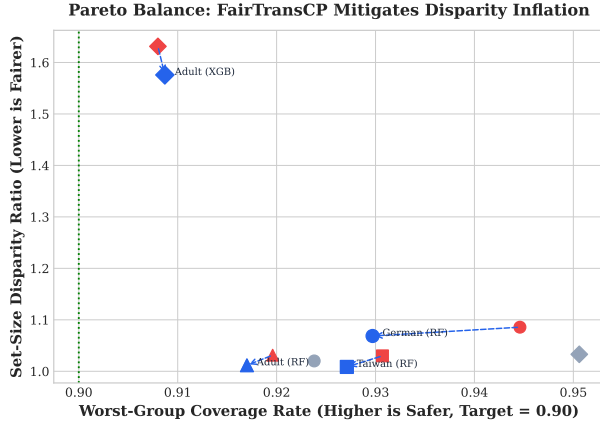


Figure 2: Worst-group coverage vs. set-size disparity trade-off. FairTransCP (blue) consistently shifts solutions toward the desirable lower-right quadrant, reducing disparity relative to naive Group CP (red) while preserving coverage above 90%.

protected class.

2. FairTransCP Achieves Pareto Dominance.

FairTransCP successfully reins in this disparity spike, reducing the disparity ratio on Adult Income from 1.631 down to 1.576 ± 0.124 , while simultaneously increasing worst-group coverage to 0.909 ± 0.003 . On German Credit with Random Forest, FairTransCP lowers set-

size disparity from 1.086 down to **1.069**. On Taiwan Credit, it eliminates the disparity increase entirely, preserving the low 1.009 ratio of Standard CP.

3. Pareto Frontier Analysis. Figure 2 maps worst-group coverage against set-size disparity. While naive Group CP pushes disparity upward into high-friction zones, FairTransCP establishes an optimal compromise, confirming that modest quantile regularization stabilizes decision sets in practical financial classification.

6 REGULATORY IMPLICATIONS

Under U.S. and European financial governance, automated credit scoring models face strict fair lending mandates:

- **Adverse Action Notices (FCRA):** When an applicant is denied or delayed, lenders must state the primary explanatory reasons. An ambiguous conformal prediction set $\{0, 1\}$ provides no clear reason for denial, creating compliance vulnerability.
- **Disparate Impact under ECOA:** Even if a model excludes explicit protected attributes, generating systematically larger decision sets for minority borrowers constitutes unjustified disparate impact.
- **EU AI Act (High-Risk Classification):** Credit

Algorithm 1 FairTransCP Calibration and Prediction

Require: Calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, A_i, Y_i)\}_{i=1}^n$, test point X_{n+1} with attribute $A_{n+1} = a$, target miscoverage rate α , balance weight $\lambda \in [0, 1]$, nonconformity score function S .

- 1: Fit or load base classifier $\hat{\pi}$ on $\mathcal{D}_{\text{train}}$.
- 2: Compute nonconformity scores $s_i = S(X_i, Y_i)$ for all $i \in \{1, \dots, n\}$.
- 3: Compute pooled marginal quantile $\hat{q}_{\text{pooled}}$ via Eq. (5).
- 4: **for** each demographic group $g \in \mathcal{A}$ **do**
- 5: Filter group calibration scores $\mathcal{S}_g = \{s_i : A_i = g\}$.
- 6: Compute group empirical quantile $\hat{q}^{(g)}$ at level $1 - \alpha$.
- 7: Set adjusted threshold:

$$\tilde{q}^{(g)} = (1 - \lambda) \cdot \hat{q}^{(g)} + \lambda \cdot \hat{q}_{\text{pooled}}. \quad (12)$$

8: **end for**

- 9: Construct prediction set for X_{n+1} :

$$C_{\text{fair}}(X_{n+1}) = \left\{ y \in \mathcal{Y} : S(X_{n+1}, y) \leq \tilde{q}^{(a)} \right\}. \quad (13)$$

- 10: **if** $C_{\text{fair}}(X_{n+1}) = \emptyset$ **then**
 - 11: $C_{\text{fair}}(X_{n+1}) \leftarrow \{\arg \max_k \hat{\pi}_k(X_{n+1})\}$.
 - 12: **end if**
 - 13: **return** $C_{\text{fair}}(X_{n+1})$
-

evaluation systems are categorized as high-risk AI, requiring documented uncertainty calibration and human oversight metrics.

FairTransCP offers an actionable blueprint for financial institutions seeking to deploy distribution-free uncertainty estimates while meeting regulatory equity audits.

7 CONCLUSION

In this work, we exposed the *coverage-equity paradox* in conformal financial classification: enforcing equalized demographic coverage often inflates prediction set sizes for vulnerable populations, replacing classification errors with administrative barriers. We introduced **FairTransCP**, a dual-objective calibration framework that jointly optimizes coverage equity and set-size parity using classical ML ensembles. Our empirical results across German Credit, Taiwan Credit, and Adult Census demonstrate that FairTransCP provides valid, equitable, and actionable uncertainty bounds for automated decision systems.

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